

SRMS-002 — The Base–Delta Asymmetry Principle

Structural Recognition above Metric Similarity (SRMS)

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Repository: <https://github.com/sizhet/Structural-Recognition-above-Metric-Similarity>

Abstract

One of the fundamental assumptions behind metric similarity is that small differences usually imply small semantic changes.

This assumption works well for many statistical learning tasks but breaks down in numerous engineering, scientific, legal, medical, and runtime systems.

This paper introduces the **Base–Delta Asymmetry Principle**, which states that a large structural base and a small structural delta do not contribute equally to meaning.

Instead, tiny structural differences frequently determine the entire semantic interpretation, runtime behavior, or engineering decision.

Recognizing these decisive structural deltas is therefore a prerequisite for reliable AI reasoning.

1. The Symmetry Assumption Behind Metric Similarity

Most similarity metrics implicitly assume that every component contributes proportionally to the overall distance.

If two objects share 99.9% of their information, they are expected to represent nearly identical meanings.

This assumption is often reasonable for statistical retrieval.

However, many engineering problems violate this assumption.

The missing 0.1% may determine everything.

2. Base and Delta Do Not Contribute Equally

Consider two software programs.

Almost every line of code may be identical.

Only one comparison operator differs.

Program A

if (temperature > 80)

Program B

if (temperature >= 80)

The structural difference is extremely small.

The runtime behavior may be completely different.

The majority of the program forms the structural base.

The comparison operator forms the decisive structural delta.

Their contributions are highly asymmetric.

3. The Base–Delta Asymmetry Principle

We therefore propose the following principle.

Large structural bases provide context. Small structural deltas frequently determine meaning.

The structural base establishes:

- context
- function
- runtime environment
- semantic scope

The structural delta determines:

- final interpretation
- decision boundary
- execution path
- runtime behavior
- engineering outcome

Meaning is therefore not distributed uniformly across all structures.

4. Small Delta, Large Consequence

Many real-world systems exhibit this asymmetry.

Examples include:

Software Engineering

==

!=

>

>=

return true

return false

One symbol changes.

The entire execution path changes.

Safety Systems

SAFE

UNSAFE

A single missing safety condition may invalidate the entire system.

Medical Diagnosis

Benign

Malignant

One critical observation changes the diagnosis completely.

Legal Documents

shall

may

One word changes legal responsibility.

Security Systems

authenticated

unauthenticated

One verification step determines whether access is granted.

These examples appear in very different domains.

Their structural pattern is identical.

A tiny delta determines the final outcome.

5. Similarity Cannot Predict Decision Boundaries

Metric similarity measures overall proximity.

Decision boundaries are determined by structural recognition.

Two objects may have:

- extremely high similarity
- completely opposite runtime behaviors

Conversely,

two objects with lower similarity may belong to the same structural category.

Recognition therefore identifies structural identity.

Similarity measures variation inside that identity.

The two serve different purposes.

6. Structural Recognition Detects the Decisive Delta

The purpose of structural recognition is not to enumerate every difference.

Its purpose is to identify which differences matter.

Recognition answers questions such as:

- Which structural element controls the decision?
- Which component changes runtime behavior?
- Which delta alters semantic identity?
- Which invariant has been violated?

These are not metric questions.

They are structural questions.

7. Why This Principle Matters for AI

Large Language Models excel at recognizing broad semantic similarity.

However, engineering tasks often depend on identifying tiny but decisive structural differences.

Examples include:

- code review
- safety verification
- scientific reasoning

- contract analysis
- software debugging
- runtime verification
- certified engineering

Future AI systems must therefore become increasingly sensitive to structural deltas rather than relying solely on global similarity scores.

8. Relation to Structural Recognition

The Base–Delta Asymmetry Principle naturally extends the central idea introduced in SRMS-001.

Structural recognition identifies:

- what structures exist,
- which structures are invariant,
- which structures determine meaning.

The Base–Delta principle explains why this recognition step cannot be replaced by similarity computation.

Similarity treats all dimensions as contributors to distance.

Structural recognition identifies which dimensions dominate semantics.

9. Engineering Implications

The principle suggests a different computational workflow.

Instead of:

Similarity

↓

Decision

Future AI systems should perform:

Structural Recognition

↓

Critical Delta Detection



Decision Boundary Analysis



Metric Similarity



Ranking

Similarity remains useful.

However, similarity should operate after the decisive structural differences have already been identified.

10. Toward Recognition-Gated Similarity

The Base–Delta Asymmetry Principle directly motivates the next step of the SRMS framework.

Rather than replacing metric similarity, future AI systems should first recognize structural identity and critical deltas. Similarity computation should then be restricted to structurally compatible candidates.

This leads naturally to the Recognition-Gated Similarity architecture introduced in the next paper.

Key Takeaways

- Large structural bases provide context rather than final meaning.
- Tiny structural deltas frequently determine semantic identity.
- Structural importance is highly asymmetric.
- Similarity cannot reliably identify engineering decision boundaries.
- Structural recognition identifies the decisive delta before similarity evaluation.
- The Base–Delta Asymmetry Principle provides the theoretical foundation

for Recognition-Gated Similarity.

SRMS Principle 002

Meaning is not uniformly distributed across structure. Large structural bases establish context, while small structural deltas frequently determine semantic identity, runtime behavior, and engineering decisions. Reliable AI must recognize decisive structural deltas before interpreting similarity.